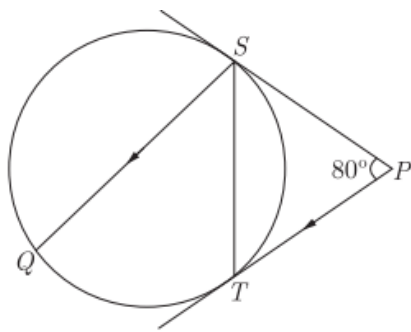


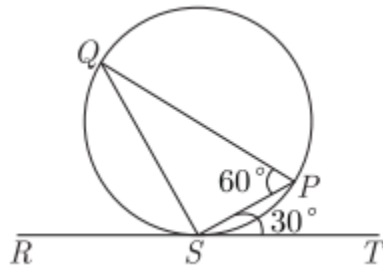
Chapter

Tangent Of Circle

1. In the given diagram, PS and PT are the tangents to the circle. Given that $SQ \parallel PT$ and $\angle SPT = 80^\circ$ The value of $\angle QST$ is: Main 2024

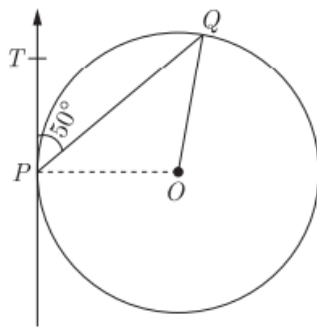


- (a) 140°
(b) 90°
(c) 80°
(d) 50°
2. In a circle with radius R, the shortest distance between two parallel tangents is equal to : SQP 2024
- (a) R
(b) 2R
(c) $2\pi R$
(d) πR
3. In the given diagram RT is a tangent touching the circle at S. If $\angle PST = 30^\circ$ and $\angle SPQ = 60^\circ$ then $\angle PSQ$ is equal to: Main 2023



- (a) 40°
- (b) 30°
- (c) 60°
- (d) 90°

4. In figure, O is the centre of circle. PQ is a chord and PT is tangent at P which makes an angle of 50° with POQ is Comp 2023

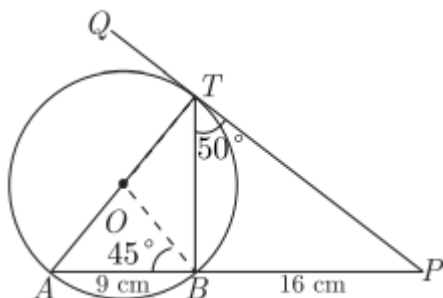


- (a) 130°
- (b) 90°
- (c) 100°
- (d) 75°

5. In the given figure, O is the centre of the circle. PQ is a tangent to the circle at T. Chord AB produced meets the tangent at P.

$AB = 9$ cm, $BP = 16$ cm, + $\angle PTB = 50^\circ$, + $\angle OBA = 45^\circ$

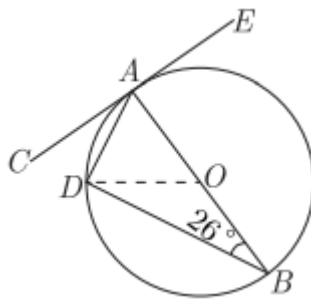
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Find:

- (i) length of PT
- (ii) $\angle BAT$
- (iii) $\angle BOT$
- (iv) $\angle ABT$

6. In the given figure O, is the centre of the circle. CE is a tangent to the circle at A. If $\angle ABD = 26^\circ$, then find Main 2023

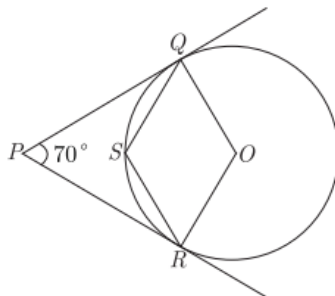


- (i) $\angle BDA$
- (ii) $\angle BAD$
- (iii) $\angle CAD$
- (iv) $\angle ODB$

7. In the given figure O is the centre of the circle. PQ and PR are tangents and $\angle QPR = 70^\circ$. Calculate.

- (i) $\angle QOR$
- (ii) $\angle QSR$

Main 2022 Sem II



Solution

1. Given:

- PS and PT are tangents
- $\angle SPT = 80^\circ$
- $SQ \parallel PT$
- Angle between two tangents = $180^\circ -$ angle subtended by chord ST at the centre
 $\Rightarrow$ Angle subtended by arc $ST = 180^\circ - 80^\circ = 100^\circ$
- Angle in the alternate segment (angle between tangent and chord) is half of this:

$$\angle QST = \frac{1}{2} \times 100^\circ = \boxed{50^\circ}$$

Correct option: (d)

2. The shortest distance between **two parallel tangents** to a circle is equal to the **diameter** of the circle.

Given radius = R,

Shortest distance = 2

Correct Answer: (b) 2R

3. Given:

- RT is a tangent at S
- $\angle PST = 30^\circ$
- $\angle SPQ = 60^\circ$

Steps:

1. **Alternate segment theorem**

Angle between tangent ST and chord SP:

$$\angle PST = \angle SQP = 30^\circ$$

2. In $\triangle PSQ$:

$$\angle SPQ = 60^\circ, \angle SQP = 30^\circ$$

3. Sum of angles in a triangle:

$$\angle PSQ = 180^\circ - (60^\circ + 30^\circ) = \boxed{90^\circ}$$

Correct Answer: (d) 90°

4.

- PT is a tangent at $P \Rightarrow OP \perp PT$

$$\angle OPT = 90^\circ$$

- Given angle between tangent PT and chord PQ is 50°

So,

$$\angle OPQ = 90^\circ - 50^\circ = 40^\circ$$

- In $\triangle OPQ$, $OP = OQ$ (radii), hence it is isosceles:

$$\angle OPQ = \angle OQP = 40^\circ$$

- Sum of angles in $\triangle OPQ$:

$$\angle POQ = 180^\circ - (40^\circ + 40^\circ) = \boxed{100^\circ}$$

Correct Answer: (c) 100°

5. Given:

- PQ is tangent at T
- $AB = 9$ cm, $BP = 16$ cm
- $\angle PTB = 50^\circ$
- $\angle OBA = 45^\circ$
- O is centre

(i) Find PT

By **tangent–secant theorem**:

$$PT^2 = PB \times PA$$

$$PA = PB + BA = 16 + 9 = 25$$

$$PT^2 = 16 \times 25 = 400 \Rightarrow \boxed{PT = 20 \text{ cm}}$$

(ii) Find $\angle BAT$

Angle between tangent and chord equals angle in the alternate segment:

$$\angle BAT = \angle PTB = \boxed{50^\circ}$$

(iii) Find $\angle BOT$

Angle at centre is twice angle at circumference on same arc:

$$\angle BOT = 2 \times \angle BAT = 2 \times 50^\circ = \boxed{100^\circ}$$

(iv) Find $\angle ABT$

- $OT \perp PT$ and $\angle PTB = 50^\circ$
 $\Rightarrow \angle OTB = 40^\circ$
- In $\triangle OBT$, $OB = OT$
 $\Rightarrow \angle OBT = 40^\circ$
- Given $\angle OBA = 45^\circ$

$$\angle ABT = 45^\circ + 40^\circ = 85^\circ$$

Final Answers:

- (i) $PT = 20 \text{ cm}$
- (ii) $\angle BAT = 50^\circ$
- (iii) $\angle BOT = 100^\circ$
- (iv) $\angle ABT = 85^\circ$

6. (i) Find $\angle BDA$

- O is the centre and the line DO is drawn to chord AB .

- The perpendicular from the centre to a chord bisects the chord and the angle subtended.
- Hence, $\angle BDA = 90^\circ$ subtends a right angle at the circumference.

$$\angle BDA = 90^\circ$$

(ii) Find $\angle BAD$

- Given: $\angle ABD = 26^\circ$
- From part (i): $\angle BDA = 90^\circ$

In $\triangle ABD$:

$$\angle BAD = 180^\circ - (26^\circ + 90^\circ) = \boxed{64^\circ}$$

(iii) Find $\angle CAD$

By alternate segment theorem,

angle between tangent and chord AD :

$$\angle CAD = \angle ABD = \boxed{26^\circ}$$

(iv) Find $\angle ODB$

- Given: $\angle ABD = 26^\circ$
- Angle at the centre is twice the angle at the circumference on the same arc:

$$\angle AOD = 2 \times 26^\circ = 52^\circ$$

- In $\triangle ODB$, $OB = OD$ (radii), so base angles are equal:

$$\angle ODB = \angle OBD = \boxed{26^\circ}$$

7. Given:

- PQ and PR are tangents from P
- $\angle QPR = 70^\circ$
- O is the centre

(i) Find $\angle QOR$

Angle between two tangents is supplementary to the angle subtended at the centre:

$$\angle QPR + \angle QOR = 180^\circ$$

$$\angle QOR = 180^\circ - 70^\circ = \boxed{110^\circ}$$

(ii) Find $\angle QSR$

Angle at the centre is twice the angle at the circumference on the same arc:

$$\angle QSR = \frac{1}{2}\angle QOR = \frac{1}{2} \times 110^\circ = \boxed{55^\circ}$$

Final Answers:

- (i) $\angle QOR = 110^\circ$
- (ii) $\angle QSR = 55^\circ$